

String methods

1. String Length

```
js  
  
let str = "Hello, World!";  
  
console.log(str.length); // 13
```

2. Changing Case

- **toUpperCase()** - Converts string to uppercase.
- **toLowerCase()** - Converts string to lowercase.

```
js  
  
console.log(str.toUpperCase()); // "HELLO, WORLD!"  
console.log(str.toLowerCase()); // "hello, world!"
```

3. Extracting Substrings

- **slice(start, end)** - Extracts part of the string.
- **substring(start, end)** - Similar to slice but does not accept negative indices.
- **substr(start, length)** - Extracts part of the string based on length.

```
js  
  
console.log(str.slice(0, 5)); // "Hello"  
console.log(str.substring(0, 5)); // "Hello"  
console.log(str.substr(7, 5)); // "World"
```

4. Replacing Strings

- **replace("old", "new")** - Replaces first occurrence.
- **replaceAll("old", "new")** - Replaces all occurrences.

```
js  
  
let text = "JavaScript is great, and JavaScript is fun!";  
  
console.log(text.replace("JavaScript", "JS")); // "JS is great, and JavaScript is fun!"  
console.log(text.replaceAll("JavaScript", "JS")); // "JS is great, and JS is fun!"
```

5. Searching in Strings

- **indexOf("text")** - Returns first occurrence index.
- **lastIndexOf("text")** - Returns last occurrence index.
- **includes("text")** - Returns true or false if found.
- **startsWith("text")** - Checks if the string starts with a given value.
- **endsWith("text")** - Checks if the string ends with a given value.

js

```
console.log(str.indexOf("World")); // 7
console.log(str.lastIndexOf("o")); // 8
console.log(str.includes("Hello")); // true
console.log(str.startsWith("Hello")); // true
console.log(str.endsWith("!")); // true
```

6. Removing Whitespace

- **trim()** - Removes spaces from both ends.
- **trimStart()** - Removes spaces from the start.
- **trimEnd()** - Removes spaces from the end.

js

CopyEdit

```
let spacedStr = " Hello, World! ";
console.log(spacedStr.trim()); // "Hello, World!"
console.log(spacedStr.trimStart()); // "Hello, World! "
console.log(spacedStr.trimEnd()); // " Hello, World!"
```

7. Splitting and Joining Strings

- **split("separator")** - Splits the string into an array.
- **join("separator")** - Joins an array into a string.

js

```
let words = "apple,banana,orange";
console.log(words.split(",")); // ["apple", "banana", "orange"]
```

```
let arr = ["Hello", "World"];
```

```
console.log(arr.join(" ")); // "Hello World"
```

8. String Repeating

- **repeat(n)** - Repeats the string n times.

js

```
console.log("Hello ".repeat(3)); // "Hello Hello Hello "
```

9. String Padding

- **padStart(length, "char")** - Pads the beginning.
- **padEnd(length, "char")** - Pads the end.

js

```
let num = "5";
```

```
console.log(num.padStart(4, "0")); // "0005"
```

```
console.log(num.padEnd(4, "0")); // "5000"
```